Escape speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain escape in an ideal gravitational-energy model.

SUCCESS CRITERIA - I can...

1. State the zero-at-infinity energy convention.
2. Distinguish escape speed from circular-orbit speed.
3. Name neglected effects in the ideal calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg cart at 10 m/s hits an identical cart at rest. Final velocities are 4 and 6 m/s. Check momentum and energy; classify the collision.

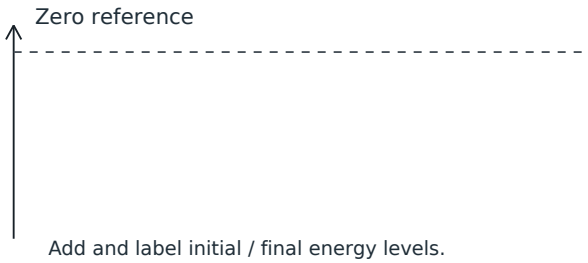
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_{escape} = \sqrt{(2\ GM/r)}$; $v_{circular} = \sqrt{(GM/r)}$. Ideal spherical body; ignore atmosphere/rotation. Minimum escape reaches infinity at zero speed with no further thrust.

Worked reasoning: For a simulated planet with $GM=50\ m^3/s^2$ and radius 4 m, escape speed is $\sqrt{(2\ GM/R)}=5$ m/s. At that limiting speed the ideal final speed at infinity is zero.

Your turn: Draw initial kinetic and gravitational-potential-energy bars whose total is zero.



Add and label initial / final energy levels.

Label K/m, U/m and total E/m in J/kg.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Does escaping mean that gravity turns off at the launch point? Explain what the speed allows in an energy model.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

04 Apply the physics

For $GM=400\ m^3/s^2$ and r=8 m, calculate circular speed and escape speed. Compare their ratio. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

Escape speed

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Continue down each column.

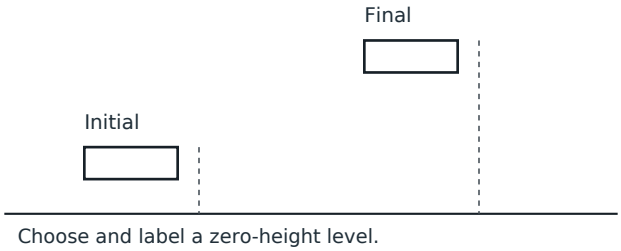
05 Explain your reasoning

Why is an ideal escape-speed calculation not a complete rocket-launch plan?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

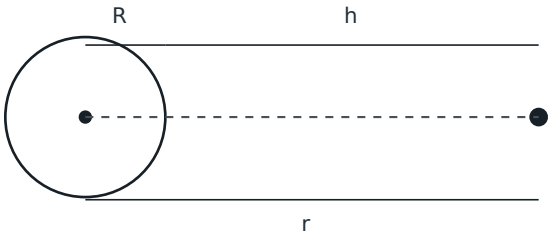
A 6 kg object descends 8 m near Earth. Use g=10. Find ΔU and the work done by gravity.
Label the distances or energy levels used in your reasoning.



Marzano check - circle: 1 2 3

07 Exit check

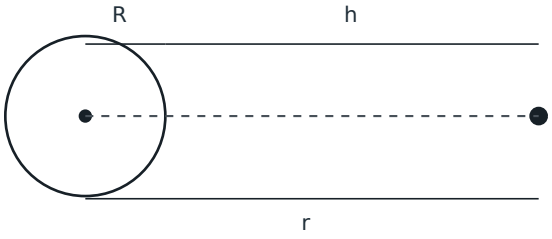
At fixed GM, launch distance increases by factor 4. Compare escape speeds. Use the energy model to show the quantities you compare; label units and any unknowns.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A planet's mass doubles while radius stays fixed. Compare escape speed. Then double both mass and radius and compare again. Use the energy model to show the quantities you compare; label units and any unknowns.



Marzano check - circle: 1 2 3